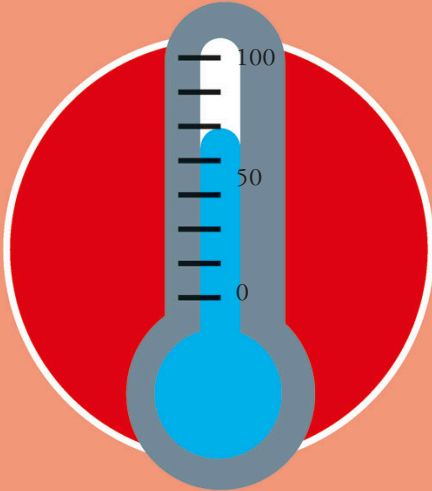




LIVING SPACE

Everything needs a certain amount of space to live and grow. How much varies enormously. Bacteria can thrive in the tiniest of spaces, while a Siberian tiger needs around 120 sq. miles (300 sq. km) of territory in which to roam. If there is not enough space, populations become overcrowded, compete for food, water, and mates, and disease spreads quickly throughout the species.



TEMPERATURE

Earth has a variety of climates ranging from very hot around the equator to freezing cold at the poles. Despite these extremes, life still manages to exist in these places. Antarctica can get as cold as -76°F (-60°C)—that's four times as cold as your freezer!—yet emperor penguins spend months incubating their eggs there and life has been found deep below the ice of Lake Vostok. At the other end of the scale, temperatures can climb to 140°F (60°C) across North African deserts where animals and plants manage to thrive.



NUTRIENTS

We all rely on these essential chemicals to build and repair body tissues, carry out vital functions, and produce energy. Animals obtain nutrients by eating food. Plants take them in from soil and the atmosphere through their roots and leaves. Bacteria absorb them directly through their cell membrane. A lack of nutrients can lead to problems—not enough vitamin C, for example, can cause a disease called scurvy in humans. So eat your fruit!



OXYGEN

All forms of animal life need oxygen. The only exceptions are some bacteria that can survive in oxygenless environments, such as inside a cow's stomach. Almost all the oxygen in the atmosphere is made by plants as part of a process known as photosynthesis. Plants take in carbon dioxide to make food and release oxygen back out into the atmosphere. So that nice flower in your window box is also helping you breathe.